DAV PROJECT

TECH LAYOFFS 2026

Group 21

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Introduction:

This dataset contains information about global tech layoffs in the first quarter of 2026. It contains information about the company affected by layoff, the number of people affected, the reason behind said layoffs and the reaction of the stock market. It contains records for 28 companies and has 26 attributes.

Objective:

The objective of this report is to analyze the causes and the consequences of layoffs in the tech industry in Q1 2026. By looking at key variables at job loss, key sectors, AI investment, geographical regions we can identify patterns emerging in driving workforce reduction in the tech sector.

About the dataset

The dataset is titled 'Tech Layoffs 2026 Tracker' and is sourced from Kaggle. It contains 28 rows and 26 columns. Some of the key columns include

Column name	Description
company	Name of company conducting layoff
layoff_date	Date of the layoff
jobs_cut	Number of jobs cut
pct_workforce	Percentage of company workforce laid off
sector	Industry sector of the company
country	Country in which company is based
simultaneous_ai_investment_bn	AI investment made alongside layoffs ($B)
reason_stated	Official reason stated for layoffs
company_revenue_2025_bn	Company revenue in 2025 (in $B)
roles_most_affected	Most commonly affected job roles

stock_change_day_pct	Stock price change on announcement day
ai_cited	Whether AI was cited as a reason

Data Cleaning

1) Duplicate values were identified and then removed This ensured each company layoff event was counted only once in the analysis.
2) Missing values (NA) were filled using the median value of each column.
3) The layoff_date column was converted from plain text to a proper datetime format using pd.to_datetime()
4) No missing values were found after the data was cleaned

```
df.drop_duplicates(inplace=True)

df['layoff_date'] = pd.to_datetime(df['layoff_date'], errors='coerce')

num_cols = ['jobs_cut', 'pct_workforce_cut', 'company_revenue_2025_bn',
            'stock_change_day_pct', 'simultaneous_ai_investment_bn']

for col in num_cols:
    df[col].fillna(df[col].median(), inplace=True)

df['ai_cited'] = df['ai_cited'].astype(str).str.strip().str.capitalize()

print("Missing values after cleaning:")
print(df.isnull().sum())
print("\nCleaning Complete")
```

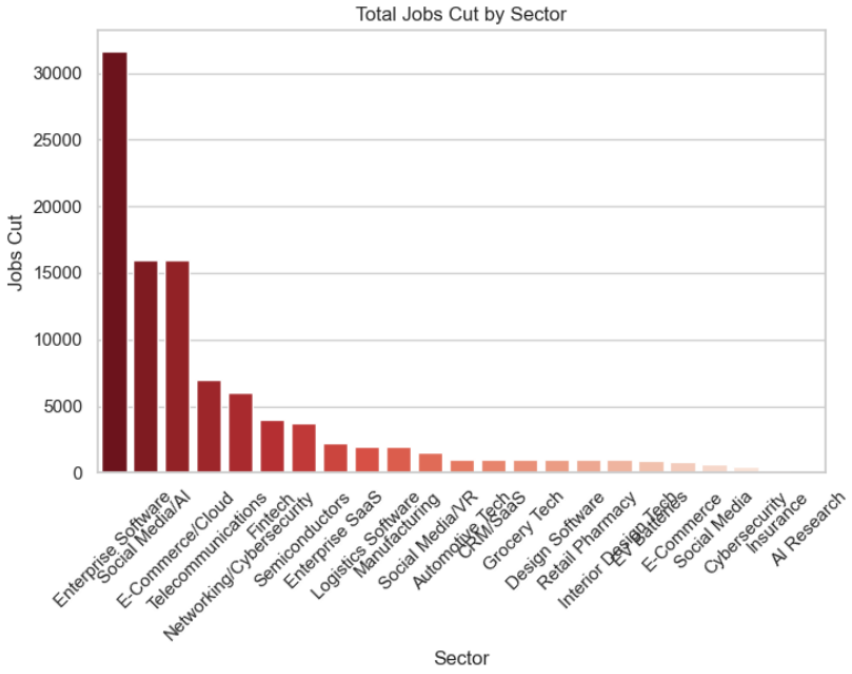
```
Missing values after cleaning:
company                          0
layoff_date                      0
jobs_cut                         0
pct_workforce_cut                0
sector                           0
country                          0
hq_city                          0
ai_cited                         0
reason_stated                    0
company_revenue_2025_bn          0
pre_layoff_headcount             0
stock_change_day_pct             0
simultaneous_ai_investment_bn    0
roles_most_affected              0
replacement_roles                0
ceo_quote                        0
layoffs_2024                     0
layoffs_2025                     0
verified_source                  0
month                            0
quarter                          0
region                           0
layoff_size_category             0
stock_reaction                   0
laid_off_vs_headcount_pct        0
data_as_of                       0
dtype: int64

Cleaning Complete
```

Data Analysis

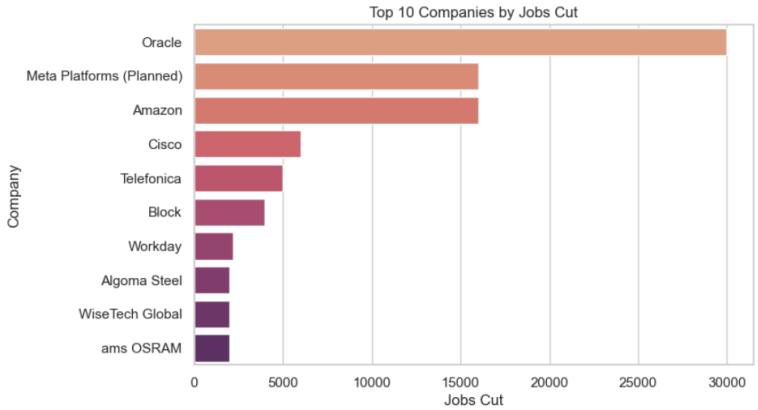
1) Total job cut by sector

Enterprise Software recorded to have the highest job cuts in 2026 (31,000) followed by social media and Ecommerce at about 15K-16K each. The remaining sectors saw fewer cuts with most below 2.5K. Large sized enterprises and tech companies were the primary drivers of layoffs in 2026



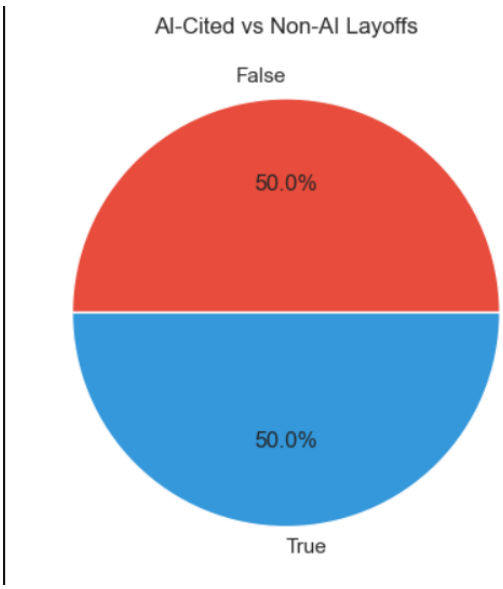
2)Top 10 Companies by Job Cuts

Oracle led with approx. 30K layoffs, more than Meta, Amazon and Cisco combined. This chart shows that layoffs in 2026 were driven by a few large corporations. The presence of non-tech firms like Telefonica and Algoma Steel show that layoffs extended beyond the tradition tech industry



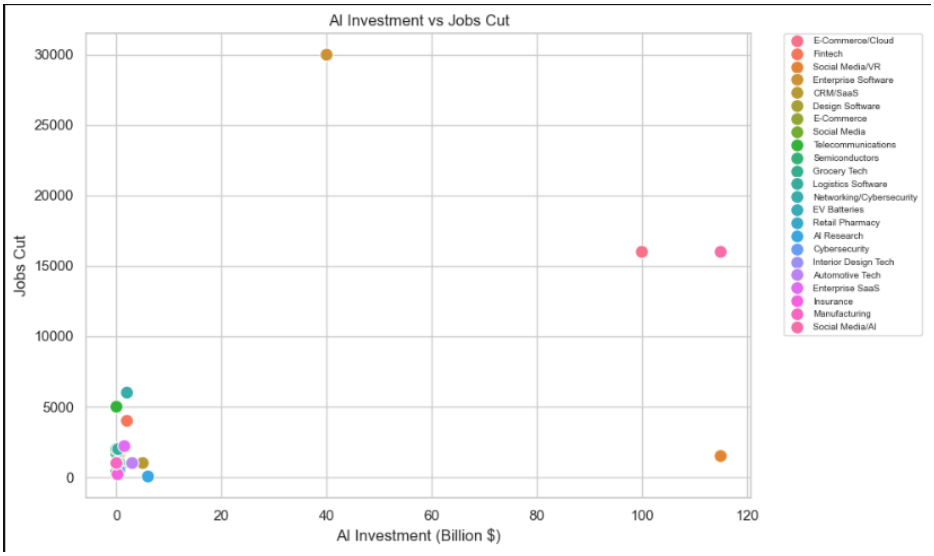
3) Is AI causing the layoffs?

This pie chart shows that exactly half of the layoffs were attributed to increase AI adoption. Hence, we can conclude that AI adoption is the primary diver of workforce reductions in tech companies in 2026.



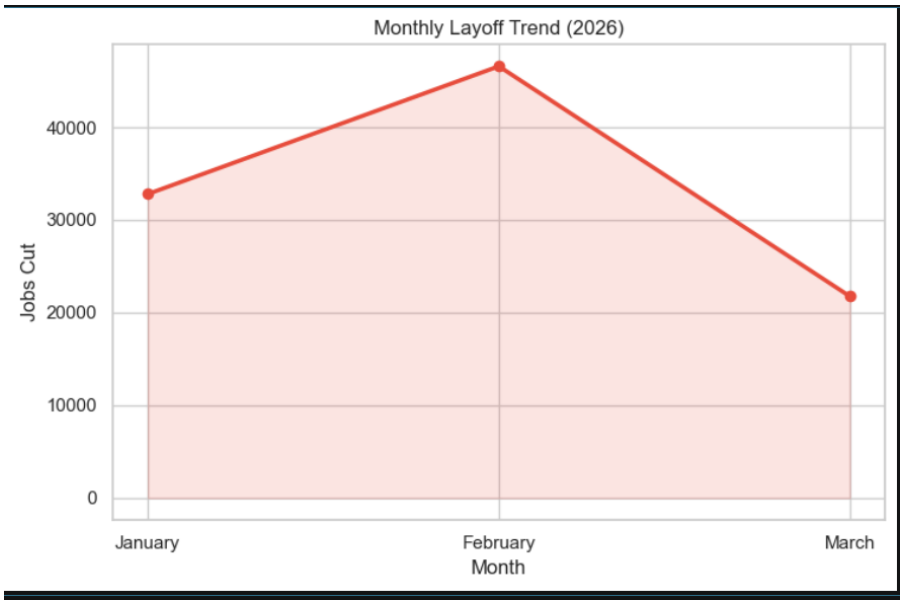
4) AI Invest vs Jobs Cuts

This scatter plot reveals that companies with near 0 AI investment still laid off people, indicating AI is not the sole reason for layoffs. E-Commerce/ Cloud stands out due to high AI spending while also cutting 16K jobs, suggesting companies channel savings from workforce reduction into AI infrastructure however other factors were also significant contributors to layoffs



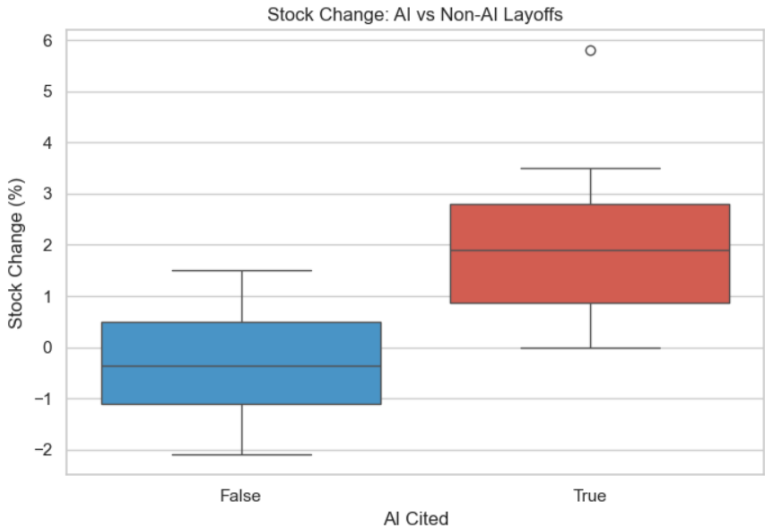
5) **Monthly layoff trend**

February 2026 was the peak month with 46,588 jobs cut, followed by January (32,800) and March (21,758). The trend shows a rise and falls pattern within Q1, aligning with the typical corporate pattern of announcing restructuring plans after Q4 earnings reviews



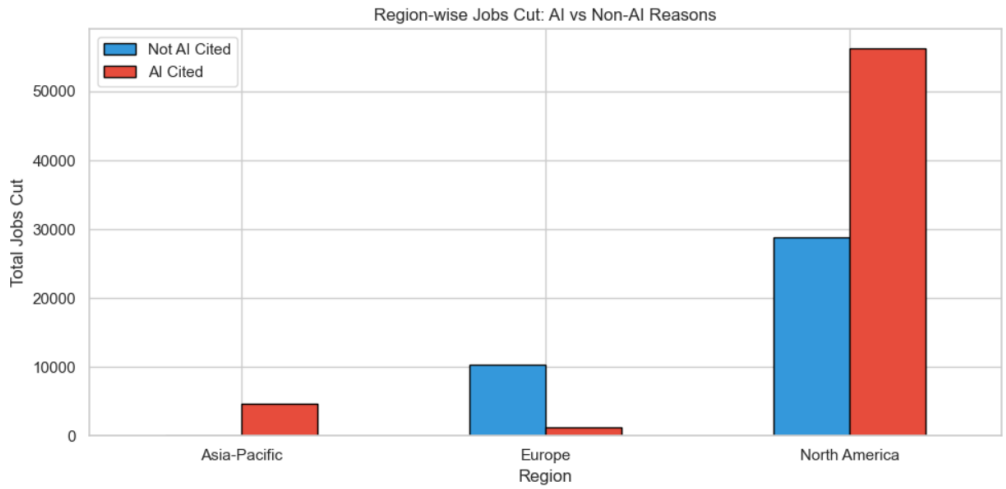
6) **Stock Market Reaction**

Out of 28 companies, 16 recorded a positive stock reaction on the day of their layoff announcement, reflecting the market's tendency to reward cost-cutting decisions regardless of the scale of job losses involved. Companies which cited AI as a reason for layoffs saw more notable positive change with a median of +2%



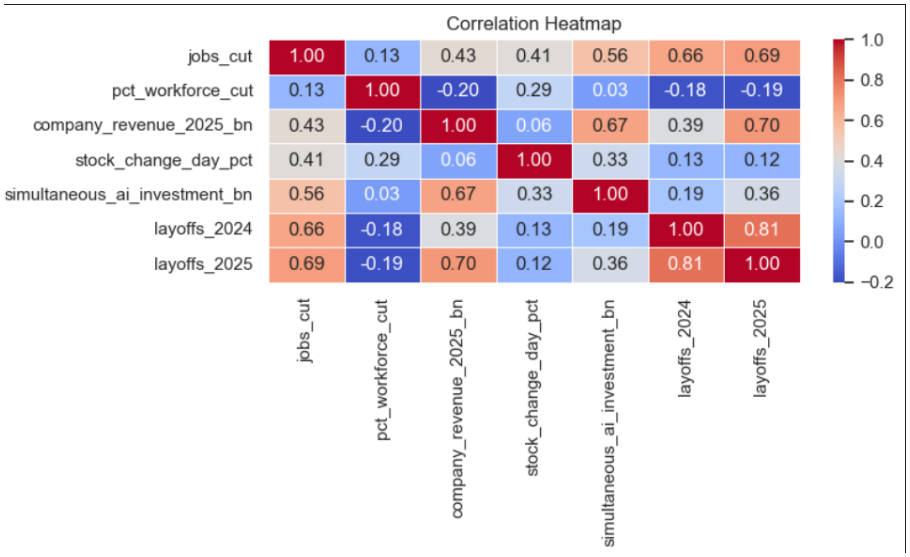
7) Layoffs by region

North America dominated total job cuts in 2026 with total layoffs being as high as ~85K and most of them being due to AI related restructuring. In contrast, Europe recorded more non-AI layoffs than AI cited one while Asia-pacific saw very little ~4.5K layoffs



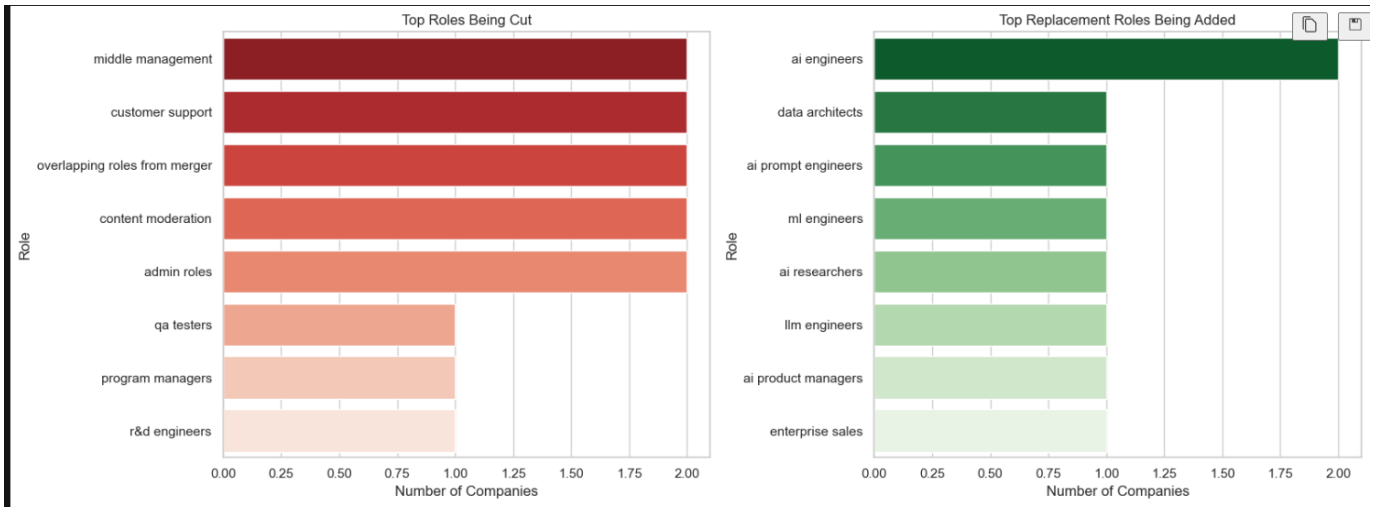
8) Correlation Heatmap

The heatmap shows AI investment has the strongest correlation with jobs cut (0.56). Layoffs in 2025 and 2026 are strongly correlated (0.81), suggesting companies that restructured before continue to do so. Company revenue also shows a moderate correlation (0.43) with jobs cut.



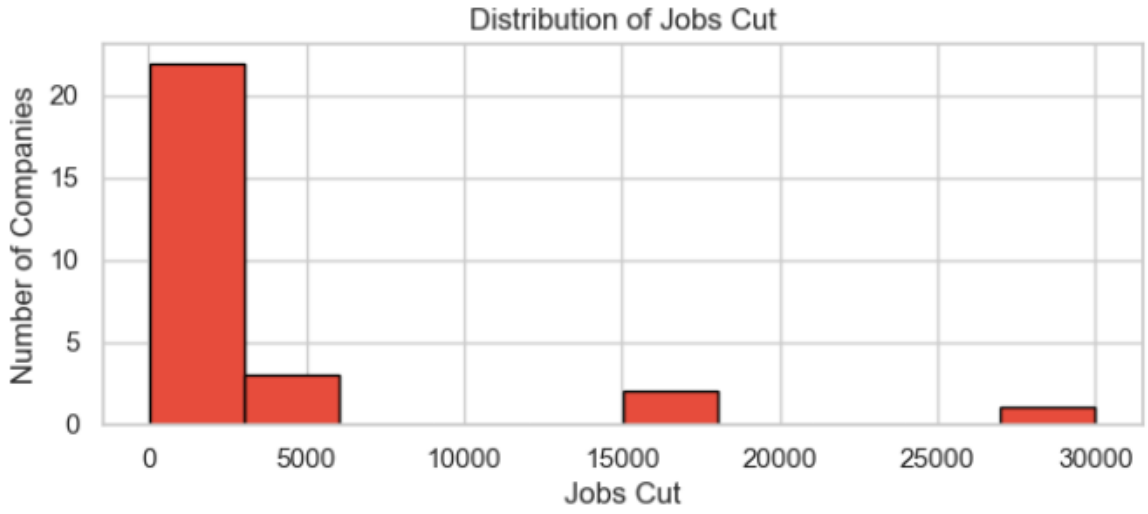
9)Roles most affected

The following data shows that companies are firing traditional operational roles which are descriptive and repetitive while simultaneously hiring AI focused talent. This is most prominent in replacing human oversight and supporting functions with AI engineers and data architects. This signals a structural shift in the workforce



10)Distribution of jobs cut

Most companies cut fewer than 1,000 jobs (median = 1,000). A few large companies like Oracle pulled the distribution heavily to the right making it right-skewed. The mean is 3,612 which is much higher than the median confirming the skewed nature.



Key insights-

1) AI is the dominant reason for layoffs. Exactly 50% of companies cited AI as a reason for workforce reductions
2) North America bears the heaviest burden. The majority of layoffs originated from the US, with companies like Oracle (30,000), Amazon (16,000), and Meta (16,000) alone accounting for a massive share of total job cuts.
3) Impact of AI on stock market - Companies citing AI as the reason largely received positive stock reactions, while non-AI layoffs (e.g., SK Battery, Walgreens, Algoma Steel) more frequently triggered negative market responses.
4) AI affects non tech industries as well- Algoma Steel, Walgreens, and SK Battery highlight that the layoff wave extended beyond pure tech into manufacturing, retail, and EV industries, driven by restructuring and market slowdowns rather than AI.
5) Non tech roles like QA testers, program managers are being cuts, proving AI disruption is moving up the skill ladder and even the jobs which are not repetitive or low skilled are unsafe
6) Companies are rebuilding their workforces around AI by hiring specialized roles such as AI/ML engineers, LLM engineers, AI researchers etc. signal that companies aren't just cutting cost but they're transforming their workforce towards being end to end AI capable

Conclusion

This analysis of the Tech Layoffs 2026 Tracker dataset reveals a clear and concerning trend in the global technology industry. A total of 101,146 jobs were cut across 27 major companies in just the first quarter of 2026, representing one of the most significant periods of tech workforce reduction in recent history. The analysis confirms that Artificial Intelligence is at the center of this transformation. Exactly 50% of all layoffs were directly attributed to AI adoption, and companies investing more in AI simultaneously cut more jobs. Oracle, the single biggest cutter with 30,000 jobs, explicitly cited AI data centers replacing human IT operations as the primary reason. Despite the human cost, the stock market largely rewarded these decisions. Companies announcing layoffs saw positive stock reactions in the majority of cases, particularly when AI was cited as the reason, with a median stock gain of 1.90% on announcement day. The findings suggest that 2026 marks a structural shift in the tech industry. AI is no longer just a tool but is actively reshaping workforce composition. Companies are replacing traditional operational roles with AI systems while hiring specialized AI

engineers, signaling a fundamental change in what human work looks like in the technology sector.

Tools Used-

The following tools were used to create this project

1. Python

Python is a high-level programming language widely used for data analysis and visualization. It provides a simple syntax and powerful libraries, making it ideal for handling large datasets and performing analytical operations efficiently.

2. Pandas

Pandas is a Python library used for data manipulation and analysis. It allows us to organize data into structures like DataFrames, making it easier to clean, filter, and analyze data. It is especially useful for handling missing values and performing statistical operations.

3. Matplotlib

Matplotlib is a visualization library used to create basic graphs such as line charts, bar graphs, and histograms. It helps in representing data visually, making it easier to understand patterns and trends.

4. Seaborn

Seaborn is an advanced data visualization library built on top of Matplotlib. It provides more attractive and informative statistical graphics, such as heatmaps, distribution plots, and scatter plots, with less code.

5. Jupyter Notebook

Jupyter Notebook is an interactive development environment used to write and execute Python code. It allows users to combine code, output, and explanations in one place, making it very useful for data analysis projects and documentation.